

Notation: $\Pr(x)$ is the input distribution and $W(y|x)$ the channel law.

1. On a discrete channel, the joint probability of sending x and receiving y is
 - (a) $\Pr(x) \Pr(y)$.
 - (b) $\Pr(x) W(y|x)$. ★

Solution: The answer is (b). The product rule of Table 1.1 gives $\Pr(x, y) = \Pr(x)W(y|x)$. Multiplying the two marginals would require independence of the input and output.

2. For arbitrary input probabilities, the MAP detector minimizes the average error probability by choosing the symbol that maximizes
 - (a) $\Pr(x) W(y|x)$. ★
 - (b) $W(y|x)$.

Solution: The answer is (a). Definition 1.3 and Theorem 1.4 give the MAP rule and its optimality. For a fixed observation, the joint probability is proportional to the posterior probability $\Pr(x|y)$.

3. For a fixed observation y , the ML detector compares
 - (a) the posterior probabilities $\Pr(x|y)$.
 - (b) the likelihoods $W(y|x)$. ★

Solution: The answer is (b). By Definition 1.3, ML maximizes $W(y|x)$ and does not use the input probabilities. MAP maximizes the posterior probability.

4. With equiprobable symbols on a Gaussian channel, both MAP and ML choose the constellation point that is
 - (a) nearest to the observation. ★
 - (b) nearest to the origin.

Solution: The answer is (a). Equal priors make MAP and ML coincide. By Proposition 1.5, the Gaussian likelihood decreases with the distance from the observation to the candidate symbol.

5. With the channel and detector fixed, changing the input probabilities leaves the conditional error probability $P_e(x)$
 - (a) unchanged. ★
 - (b) dependent on the new prior probability of x .

Solution: The answer is (a). By Definition 1.2 and the following remark, $P_e(x)$ conditions on x being sent. It depends on the channel and fixed detector, not on how often x is sent.

6. Two symbols x_0 and x_1 have probabilities q and $1 - q$. The average error probability of a detector is
 - (a) $P_e(x_0) + P_e(x_1)$.
 - (b) $qP_e(x_0) + (1 - q)P_e(x_1)$. ★

Solution: The answer is (b). Equation (1.7) weights each conditional error probability by the probability of sending that symbol.

7. A symbol X is chosen equiprobably from $\{-1, 1, -j, j\}$. Its average symbol energy E_s is
 - (a) 1. ★
 - (b) 0.

Solution: The answer is (a). Section 1.2 defines $E_s = E(|X|^2)$. Every symbol has squared magnitude 1, so $E_s = 1$. The mean symbol $E(X)$ is zero, but the average energy is not.

8. Equiprobable symbols $\pm A$, with $A > 0$, are sent over the real Gaussian channel. In terms of the SNR, the ML error probability is
 - (a) $\frac{1}{2} \operatorname{erfc}(\sqrt{2\text{SNR}})$.
 - (b) $\frac{1}{2} \operatorname{erfc}(\sqrt{\text{SNR}/2})$. ★

Solution: The answer is (b). For 2-PAM, $E_s = A^2$ and $\text{SNR} = A^2/\sigma^2$. Equation (1.22) gives $P_e = \frac{1}{2} \operatorname{erfc}(\sqrt{\text{SNR}/2})$.

9. For a fixed 2-PAM constellation on the real Gaussian channel, increasing the SNR makes the ML error probability
 - (a) increase.
 - (b) decrease. ★

Solution: The answer is (b). Equation (1.22) gives $P_e = \frac{1}{2} \operatorname{erfc}(\sqrt{\text{SNR}/2})$. The function erfc decreases with its argument, so the error probability falls as the SNR increases.

10. For equiprobable QPSK symbols $\pm a \pm ja$, with $a > 0$, on the complex Gaussian channel, an ML error when $x = a + ja$ is sent occurs if the output has
 - (a) a negative real or imaginary part. ★
 - (b) a negative real part and a negative imaginary part.

Solution: The answer is (a). In Figure 1.2(b) and Problem 1.13, the region of $a + ja$ is the first quadrant. At least one wrong sign causes an error. The axes have probability zero.